

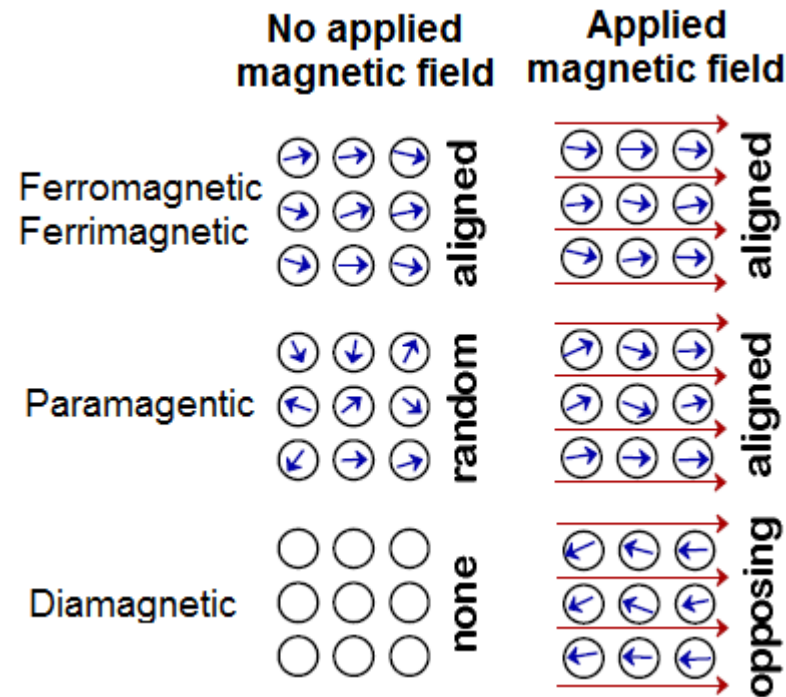
Physical Chemistry of Materials

Lecture 15

Surfaces

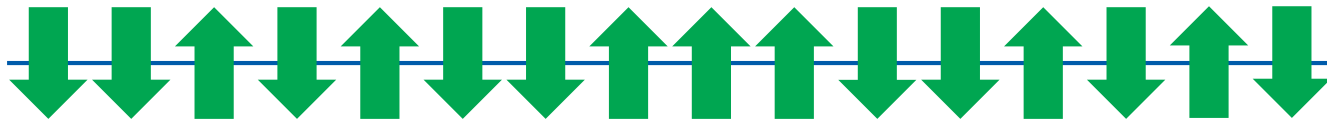


Magnetism



Independent Spin Model

Consider a solid system consisting of N spins:



$$E(\uparrow) = +\mu_B B, \quad E(\downarrow) = -\mu_B B$$

The system consists of distinguishable particles:

$$Z = z^N = [2 \cosh(\beta \mu_B B)]^N$$

(no phase transition)

Ising Model

Consideration of neighboring spin-spin interactions

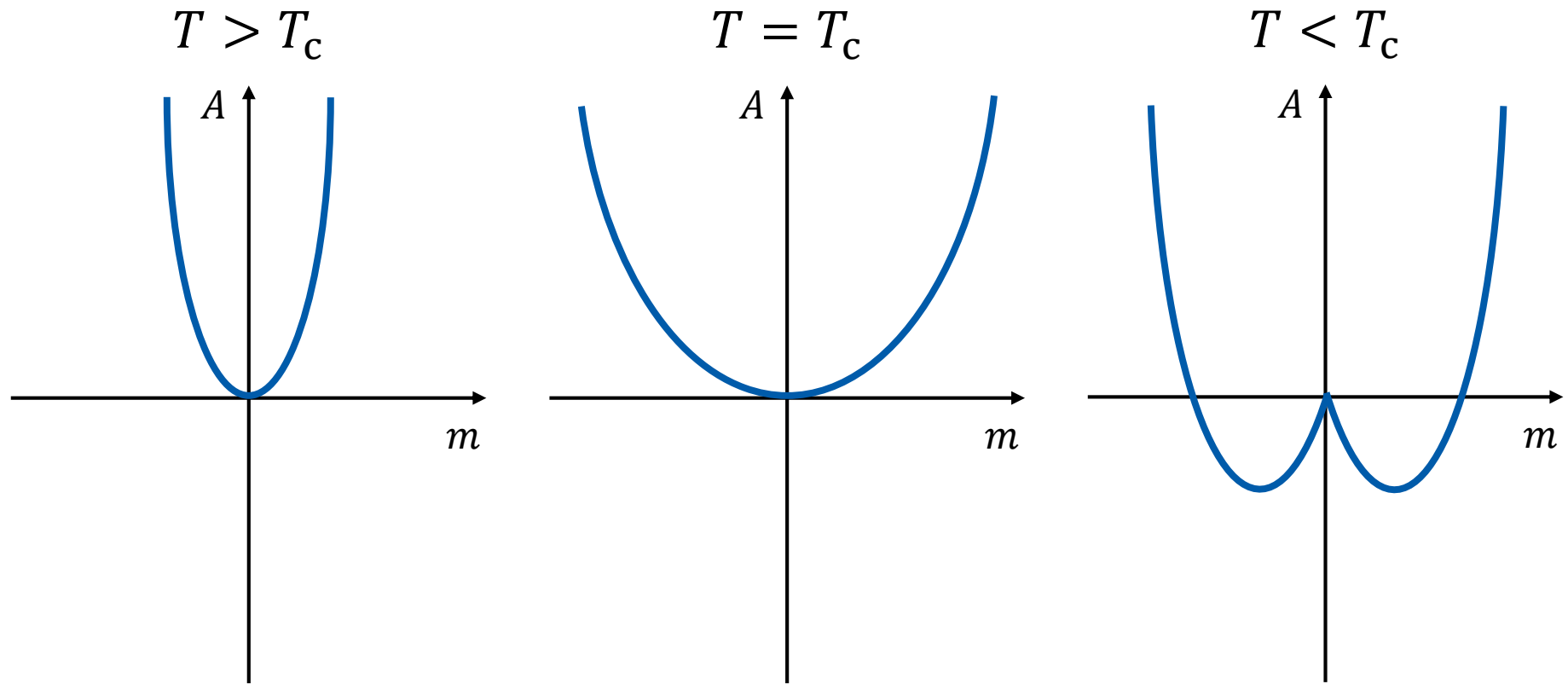
- System energy

$$E = \mu_B B \sum_i \sigma_i - J \sum_{\langle ij \rangle} \sigma_i \sigma_j$$

- Sign of J
 - $J > 0$: ferromagnetic
 - $J < 0$: antiferromagnetic

Last Class

Phase Transition in MF Model



1-Dimensional Ising Model

For $B = 0$,

$$Z = \sum_{\{\sigma_i\}} e^{\beta J \sum_{\langle ij \rangle} \sigma_i \sigma_j}$$

Consider the periodic boundary condition ($\sigma_{N+1} = \sigma_1$);

$$Z = \sum_{\sigma_1=\pm 1} \sum_{\sigma_2=\pm 1} \sum_{\sigma_3=\pm 1} \cdots \sum_{\sigma_N=\pm 1} e^{\beta J \sum_i \sigma_i \sigma_{i+1}}$$

$$Z = [2 \cosh(\beta J)]^N$$

(no phase transition)

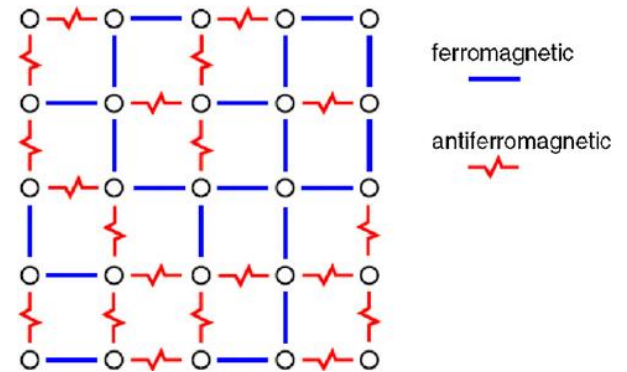
Spin Glasses

- Ising model

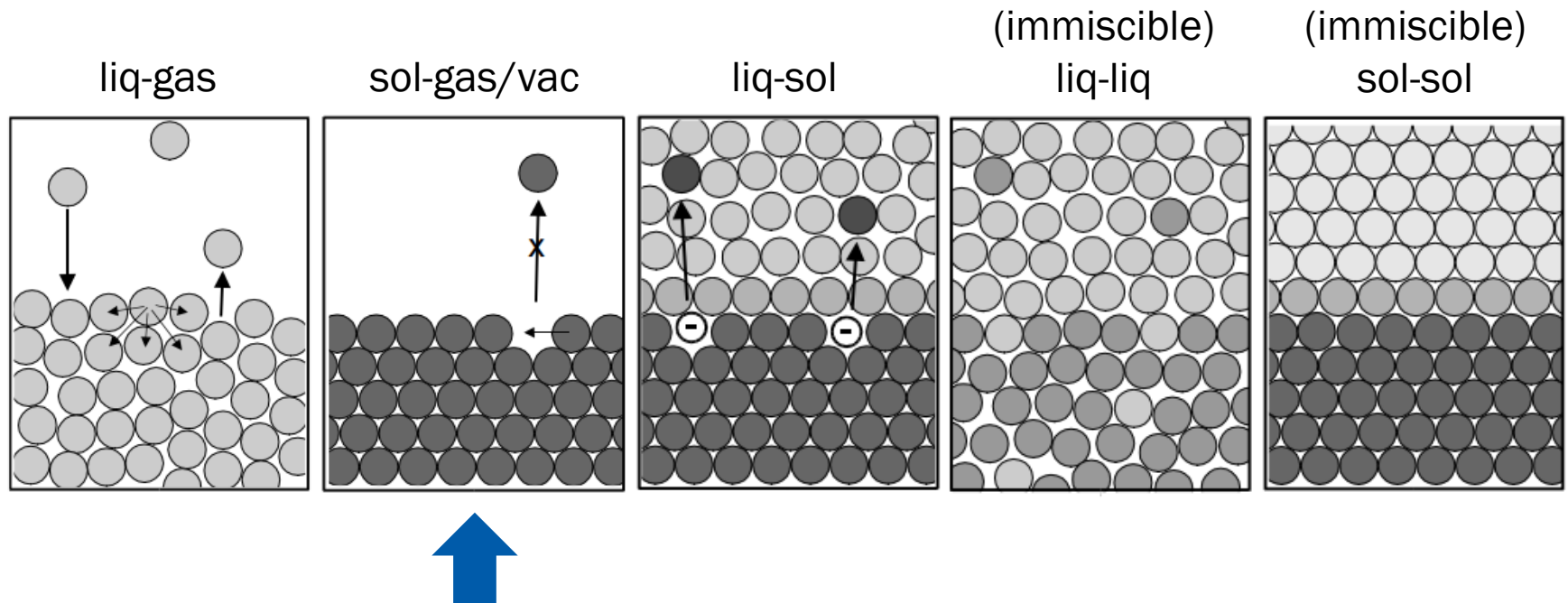
$$E = \mu_B B \sum_i \sigma_i - J \sum_{\langle ij \rangle} \sigma_i \sigma_j$$

- Edwards-Anderson model

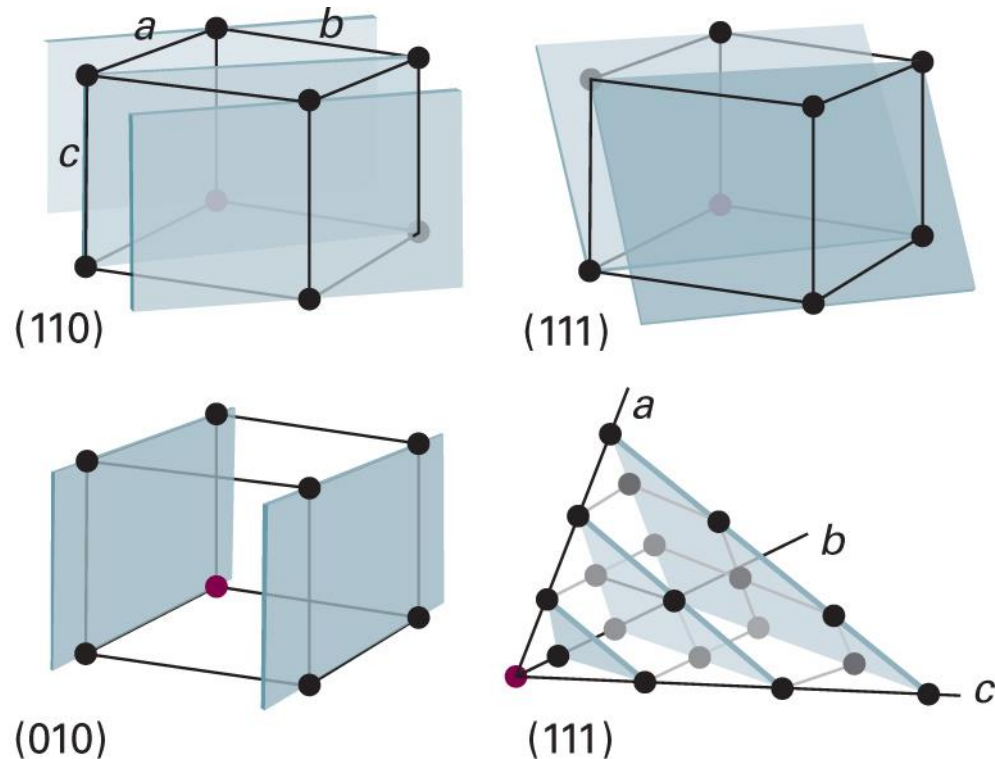
$$E = - \sum_{\langle ij \rangle} J_{ij} \sigma_i \sigma_j$$



Surfaces and Interfaces



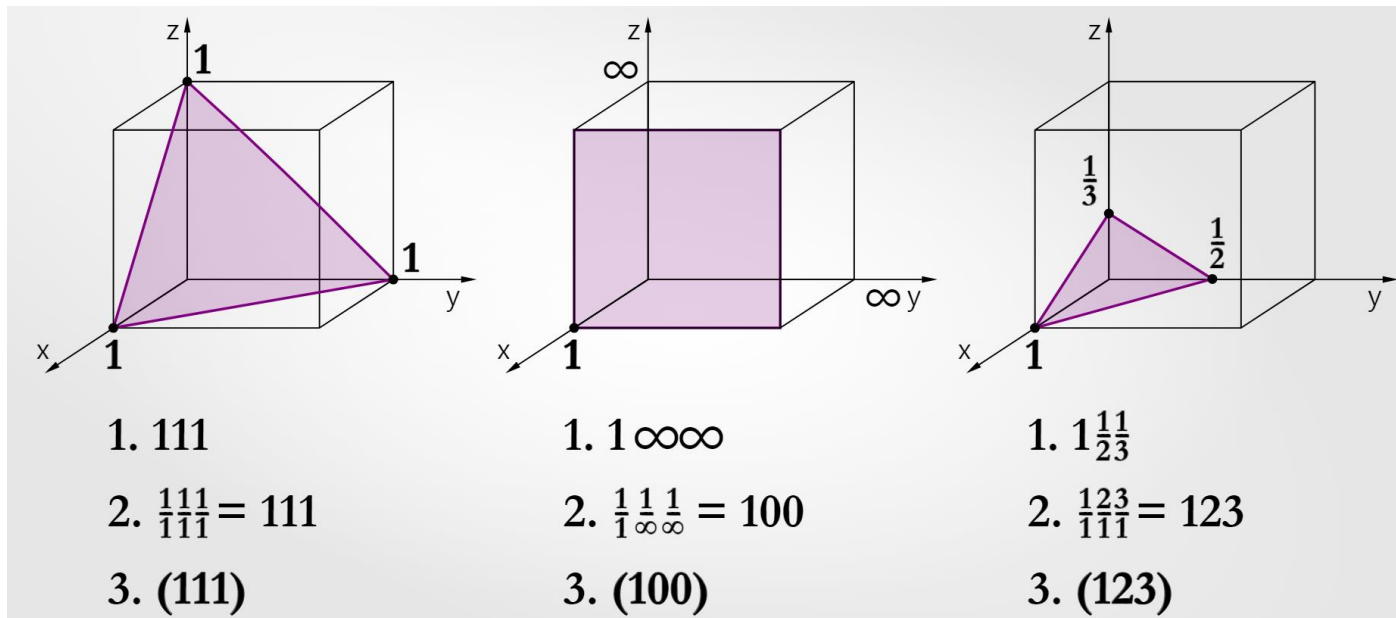
Lattice Planes



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 15A.12)

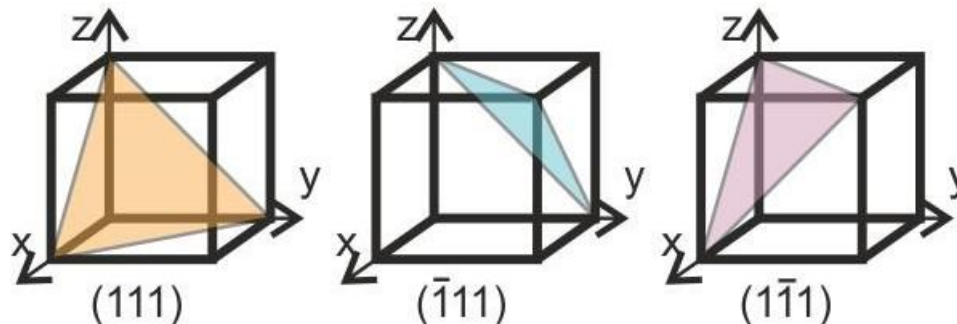
Miller Indices

Reciprocals of the intersection distances along lattice axes



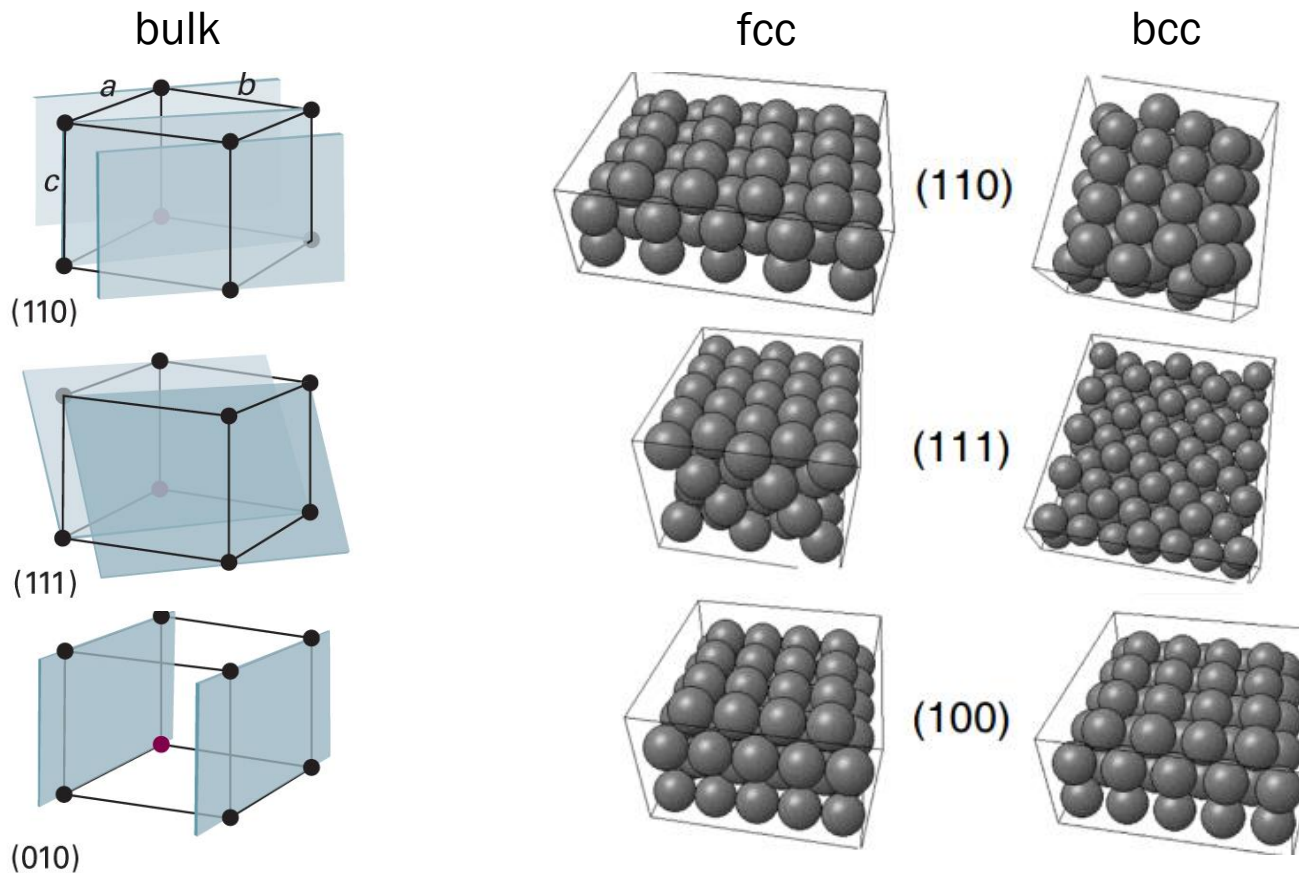
Notations

- Negative index: a bar over the number



- Brackets
 - (hkl) : individual plane
 - $\{hkl\}$: set of parallel planes

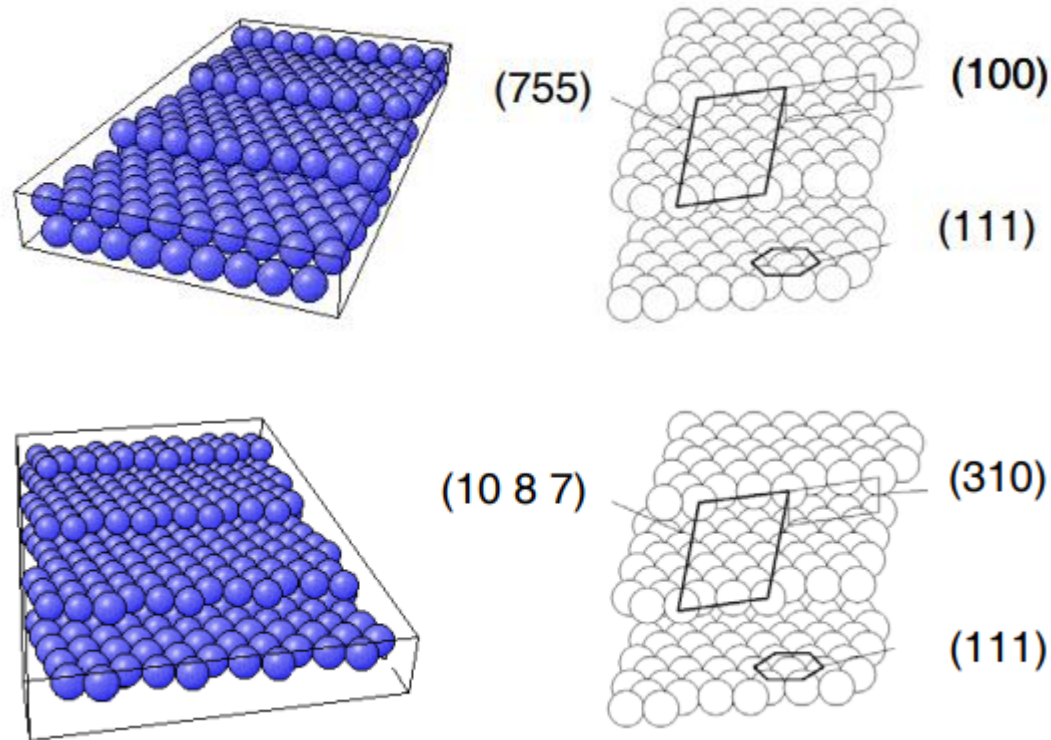
Ideal Crystal Surfaces



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 15A.11

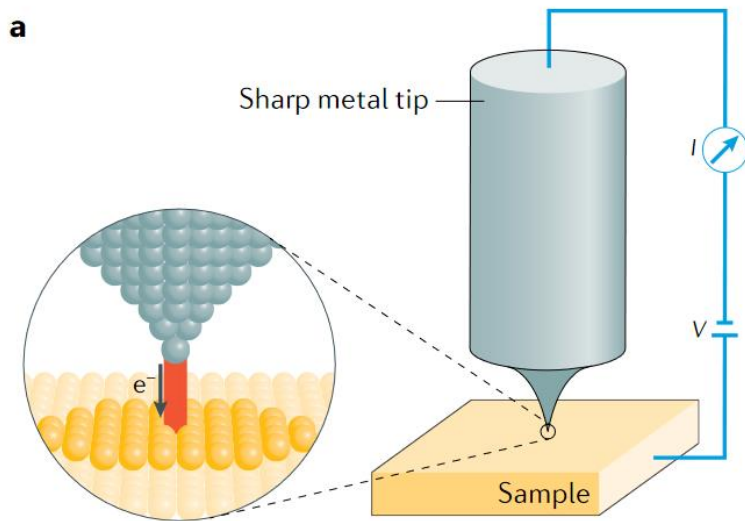
https://www.chemie-biologie.uni-siegen.de/makro/master_surf_chem/lecture_introsurfchem_1b.pdf)

High Index Surfaces

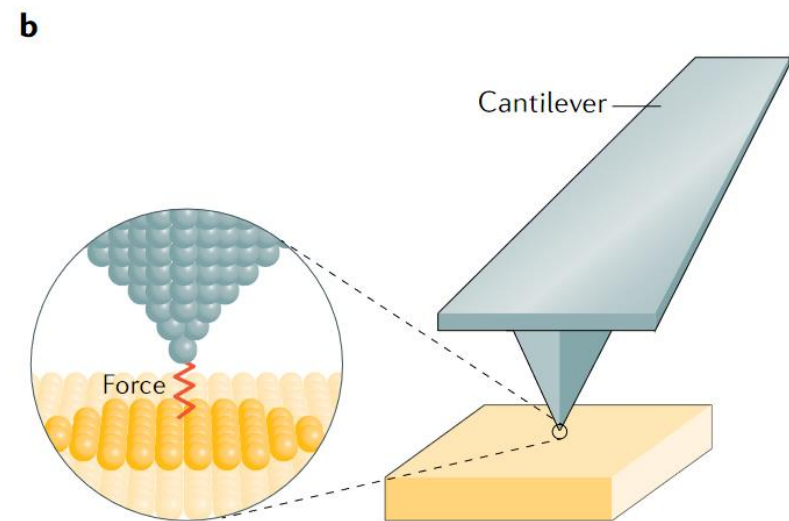


(Image: https://www.chemie-biologie.uni-siegen.de/makro/master_surf_chem/lecture_introsurfchem_1a.pdf) 13

Characterization: Microscopy

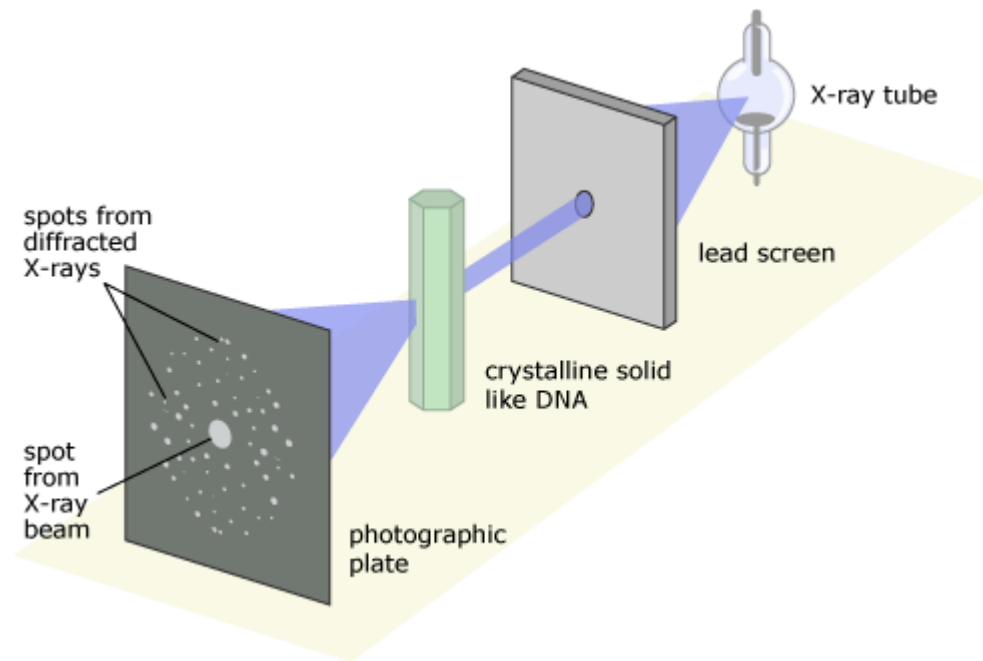


Surface Tunneling Microscope (STM)



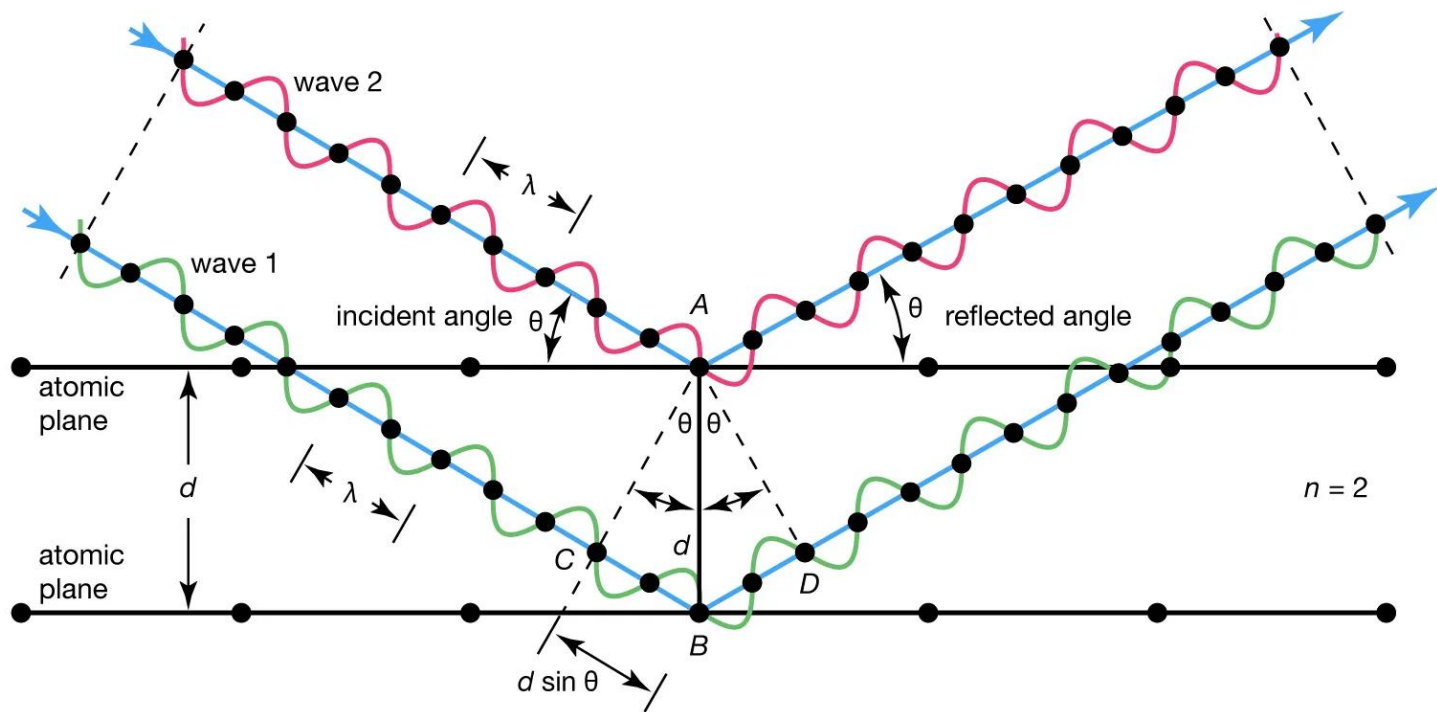
Atomic Force Microscope (AFM)

Characterization: X-ray Diffraction



(Image: <https://undsci.berkeley.edu/teach-resources/x-ray-diffraction/>)

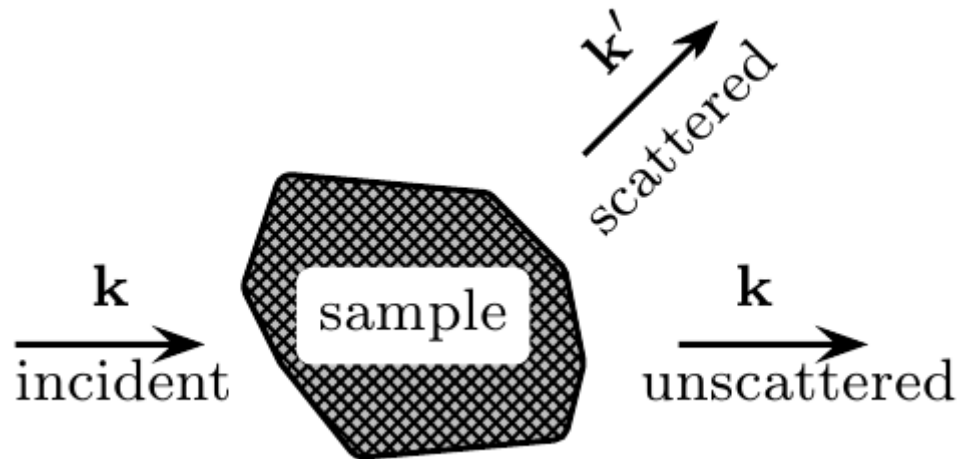
Bragg's Law



$$BC + BD = 2d \sin \theta = n\lambda$$

(Image: <https://www.britannica.com/science/X-ray-diffraction>)

Laue Equation



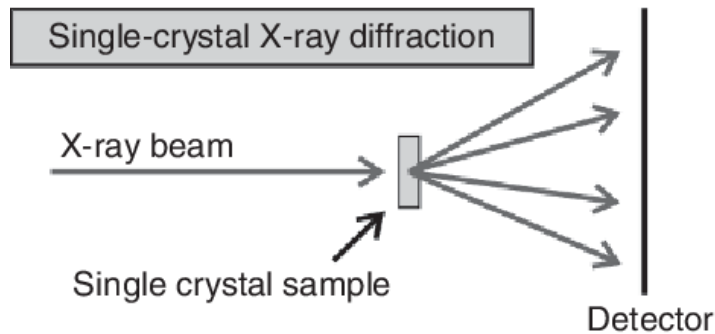
$$\vec{k}' - \vec{k} = \vec{G}$$

($\vec{G}$: lattice point in the reciprocal lattice)

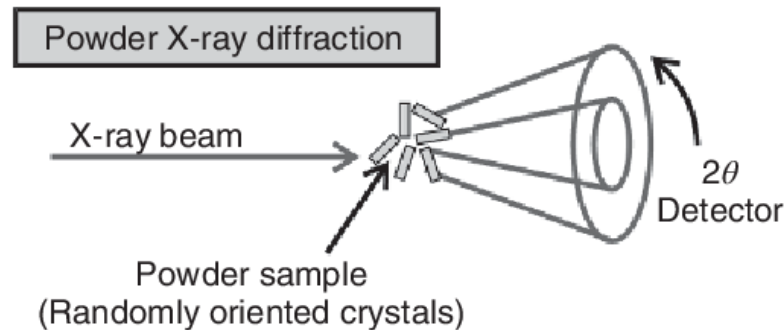
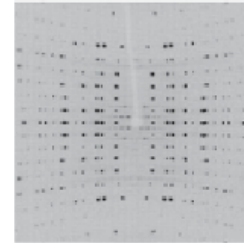
Single Crystal vs. Powder



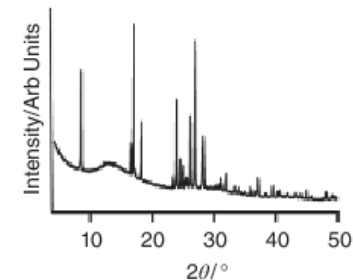
NaCl
single
crystal



Measure 3-dimensional
diffraction data



Measure 1-dimensional
diffraction data
(Peak overlap)



Electrons and Neutrons

- de Broglie wavelength

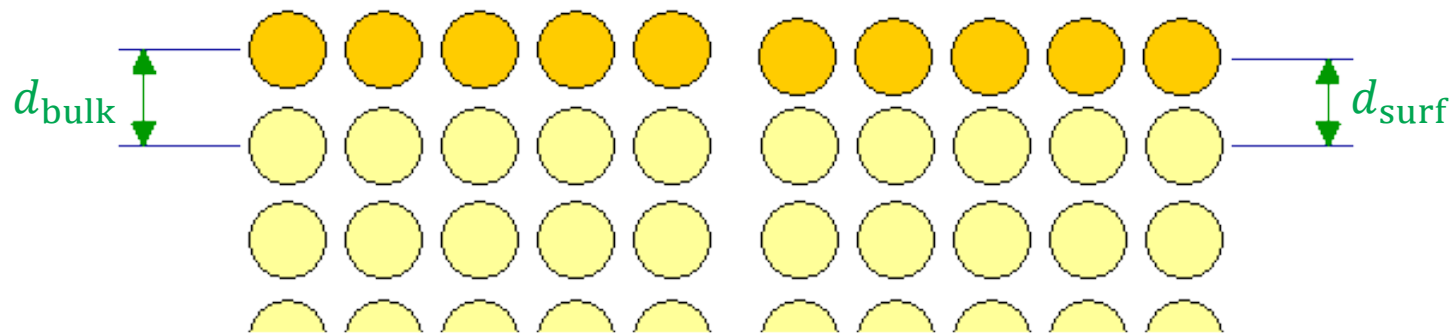
$$\lambda = \frac{h}{p}$$

X-ray	Electrons	Neutrons
No charge	Negative charge	No charge
Spin 1	Spin 1/2	Spin 1/2
Valence electrons	Core electrons Nuclei	Nuclei
Cheap	Moderate	Expensive

Real Crystal Surfaces

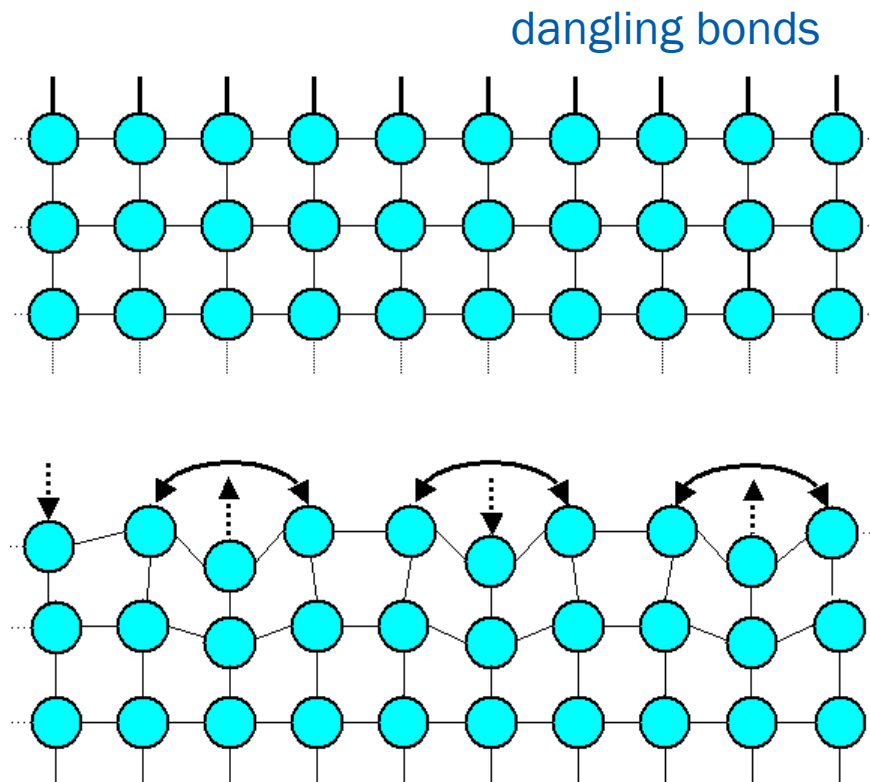
- Rearrangement
 - Relaxation
 - Reconstruction
- Defects
 - Point and line defects
 - Dislocations
 - Grain boundaries

Surface Relaxation

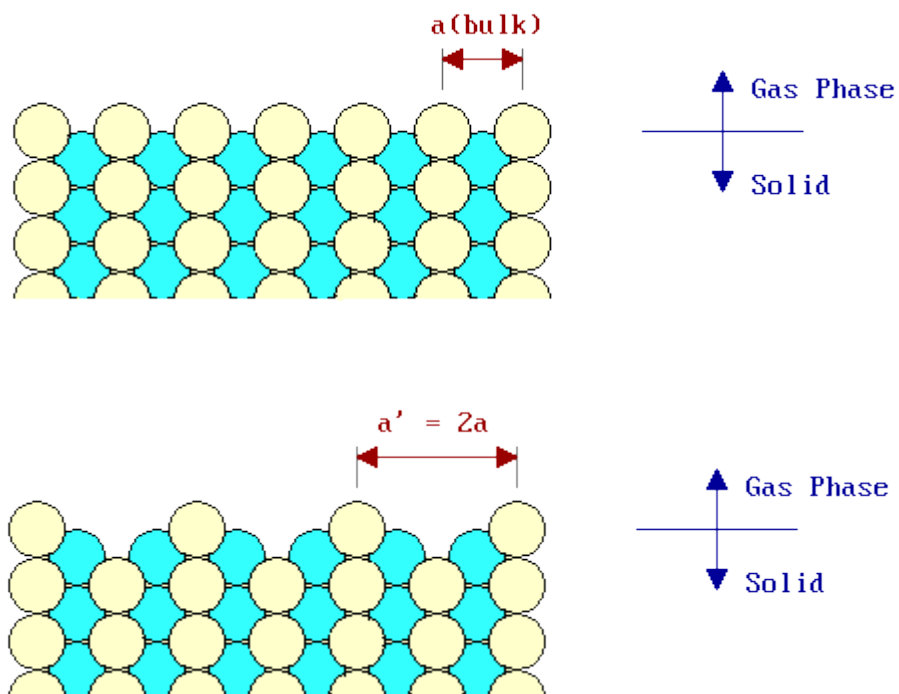


$$d_{\text{bulk}} > d_{\text{surf}}$$

Surface Reconstruction

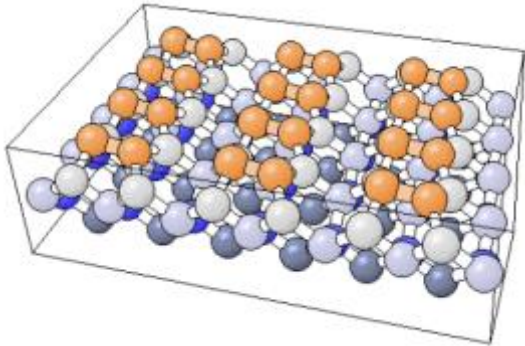


Superlattice

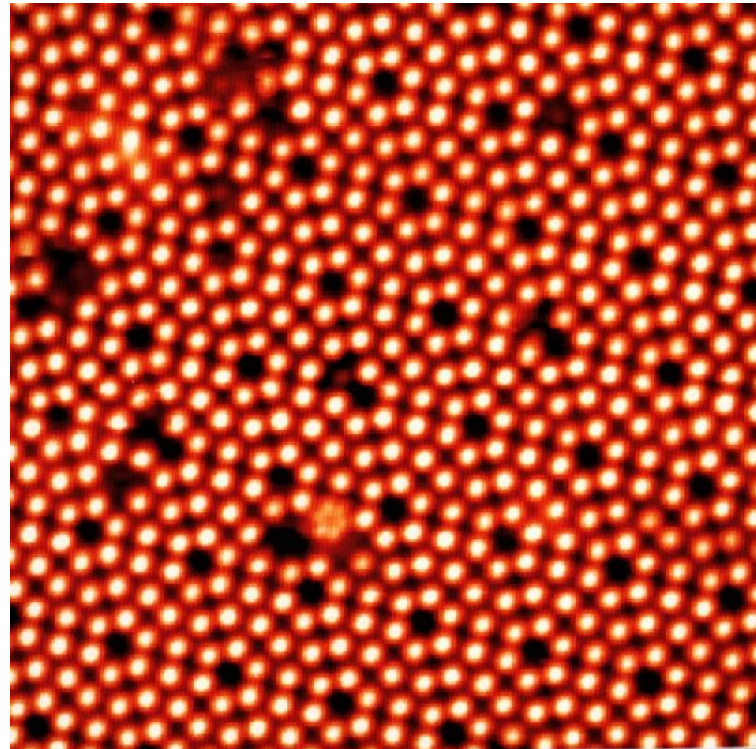


Examples

Si(100)-(2×1)



Si(111)-(7×7)

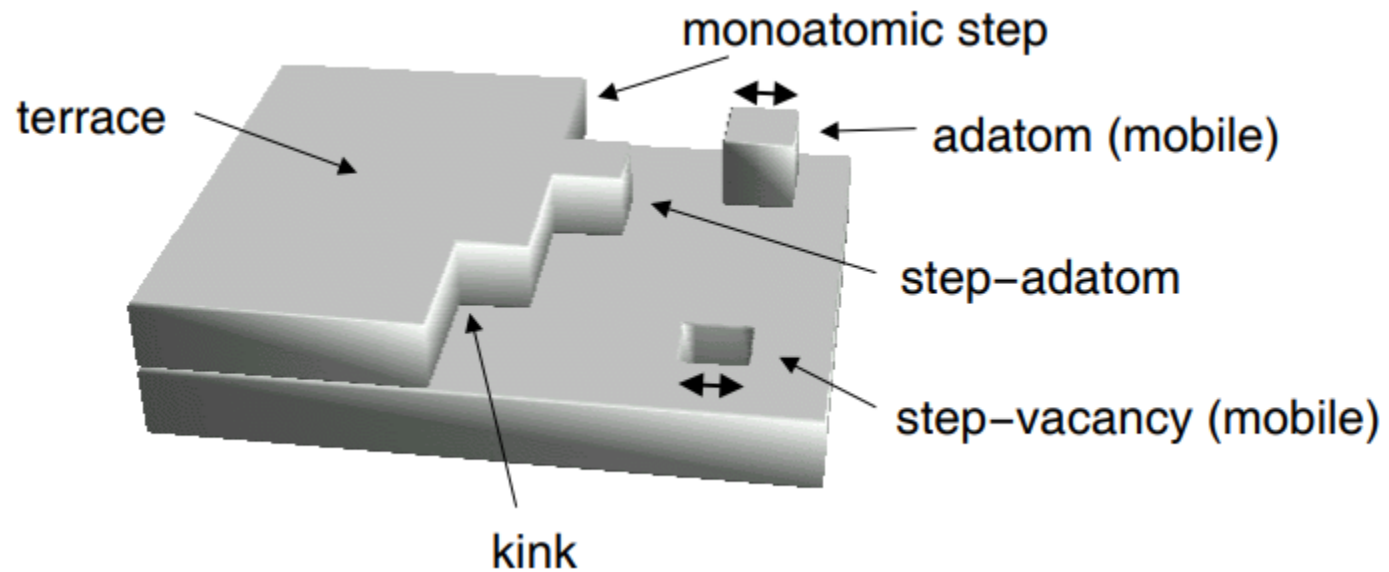


Au(111): herringbone reconstruction

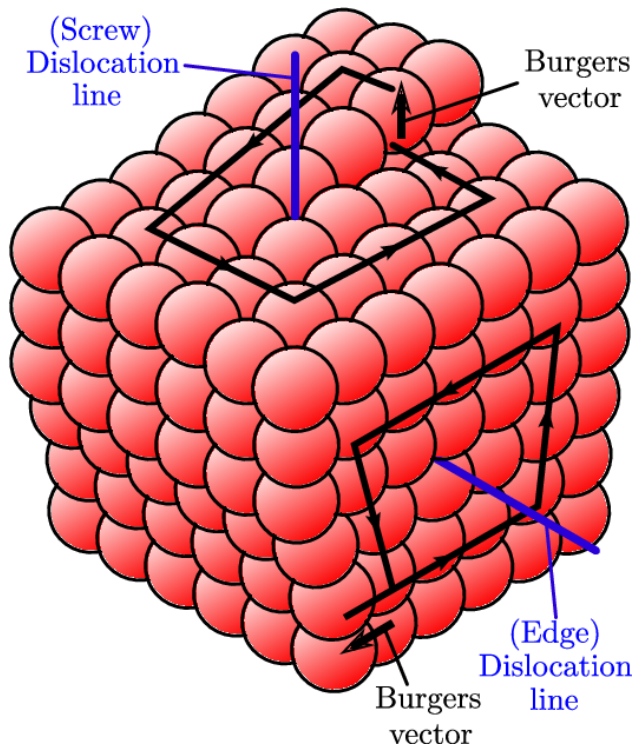


(Images: https://www.chemie-biologie.uni-siegen.de/makro/master_surf_chem/lecture_introsurfchem_1a.pdf
<https://www.physik.uni-wuerzburg.de/ep2/media/pre-2014/>)

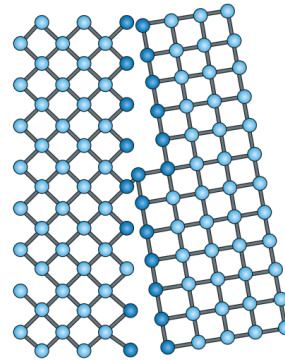
Surface Defects



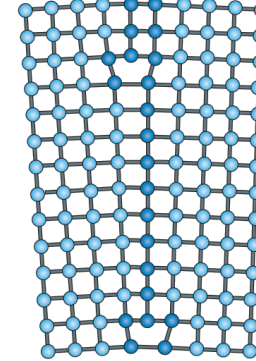
Dislocations & Grain Boundaries



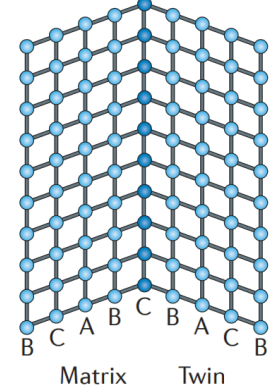
High-angle grain boundary



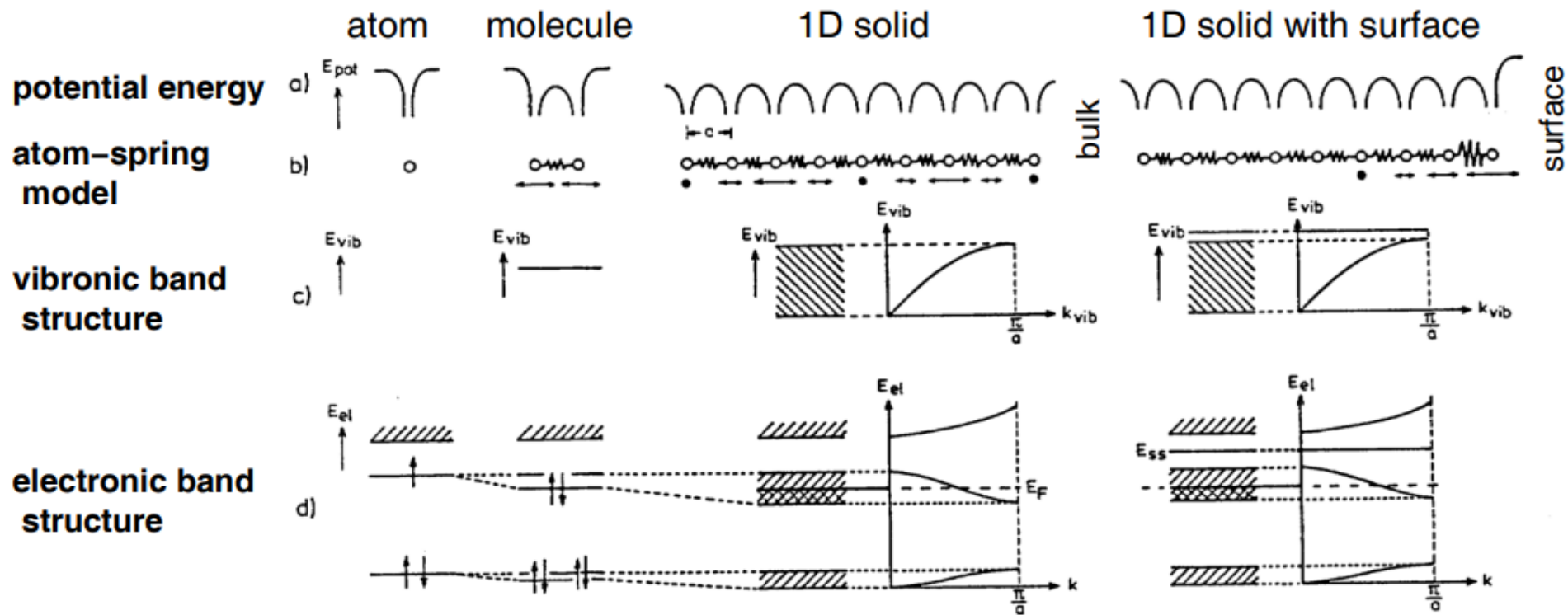
Low-angle grain boundary



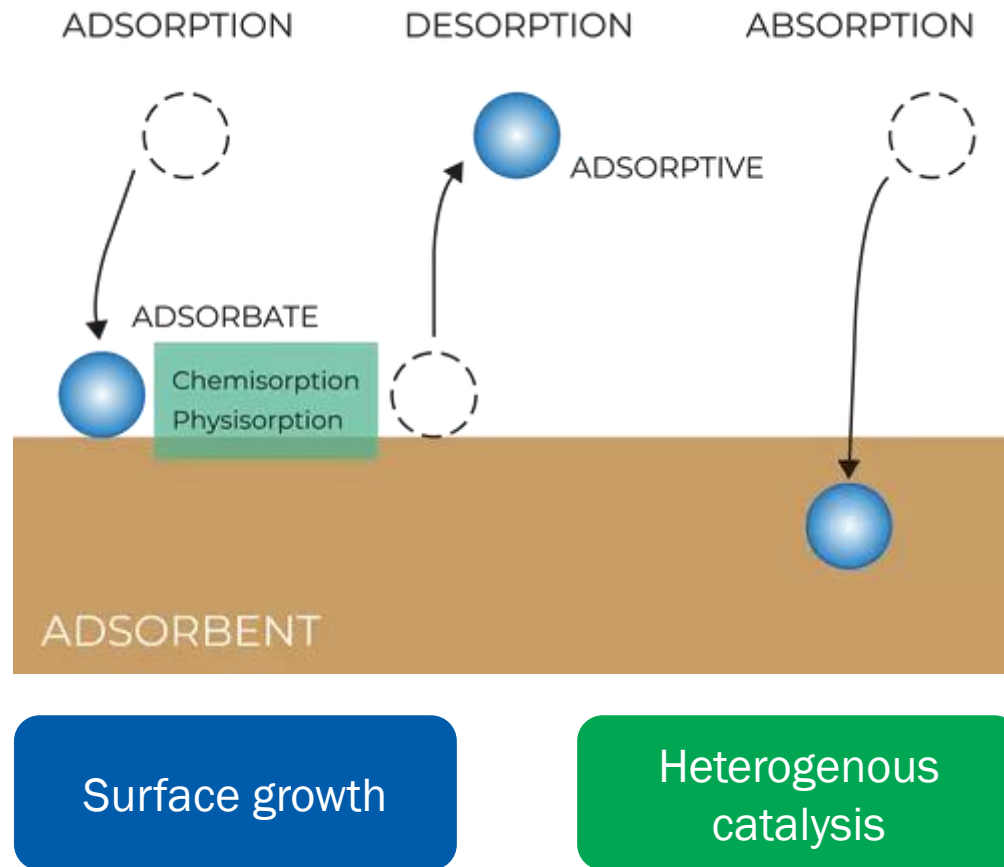
Twin boundary



Surface Effects on Band Structures

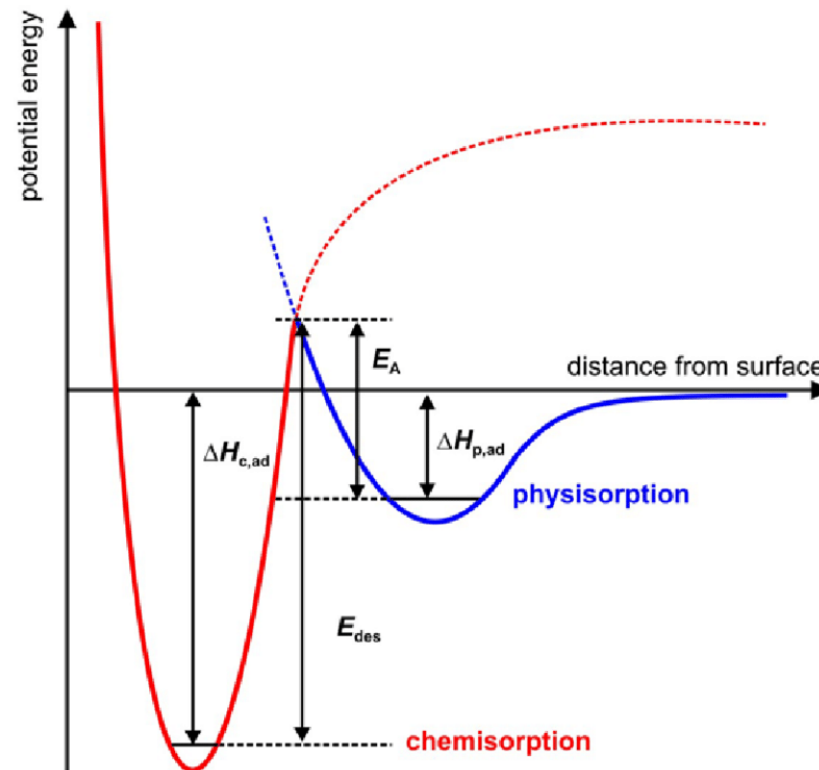


Adsorption: Gate to Surface Reaction



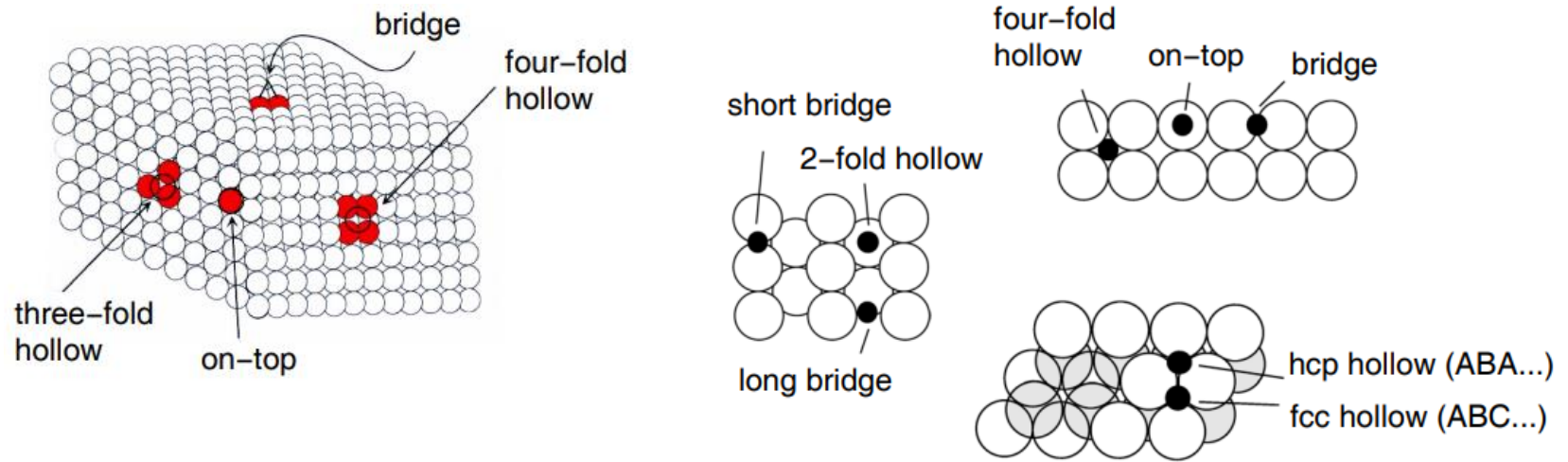
(Image: <https://www.microtrac.com/applications/knowledge-base/what-is-adsorption/>)

Physisorption vs. Chemisorption



(Image: Schwiedernoch, PhD thesis (2005))

Adsorption Sites

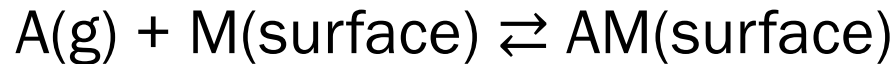


Fractional Coverage

$$\theta = \frac{\text{number of adsorption sites occupied}}{\text{number of adsorption sites available}}$$

can be used to study [adsorption kinetics](#)

Langmuir Isotherm



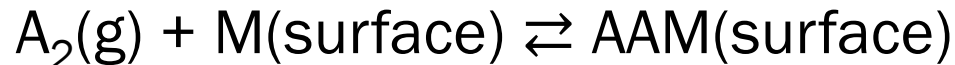
- Number of sites N & partial pressure p
- Rate law

$$\frac{d\theta}{dt} = k_a p N (1 - \theta) - k_d N \theta$$

- Steady-state solution ($\alpha = k_a/k_d$)

$$\theta = \frac{\alpha p}{1 + \alpha p}$$

Adsorption with Dissociation



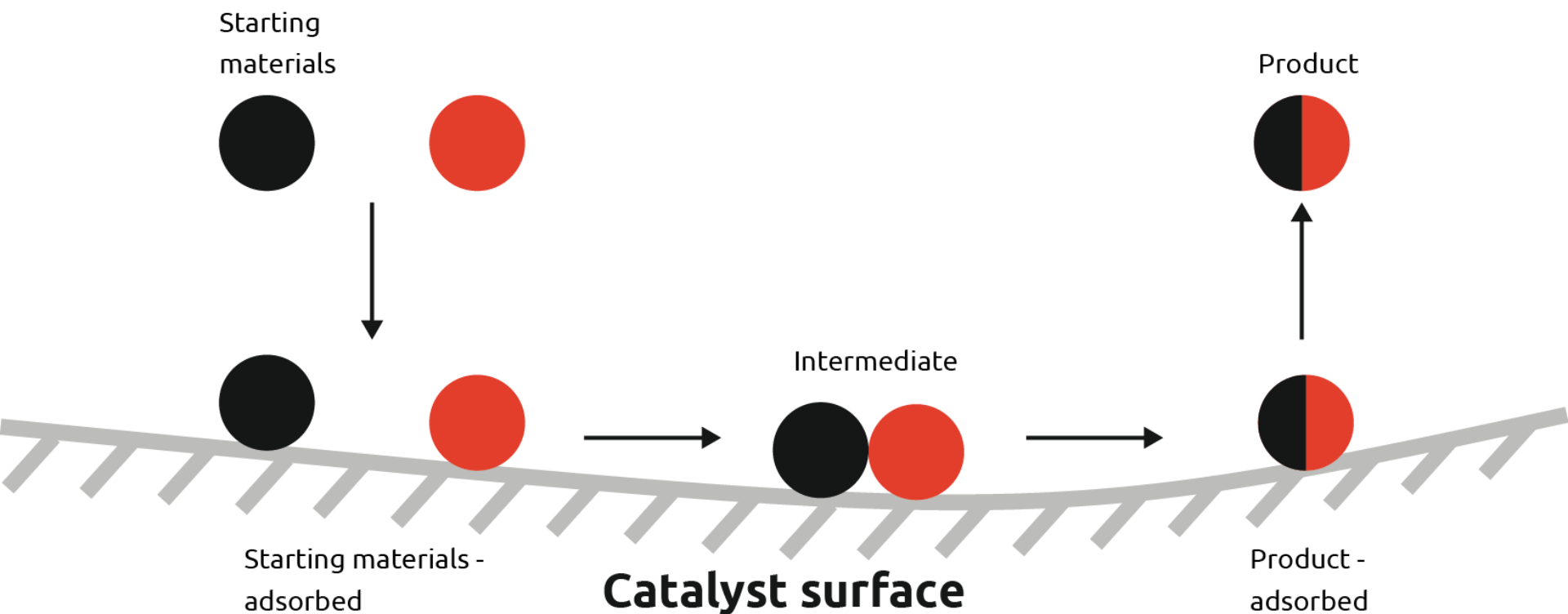
- Number of sites N & partial pressure p
- Rate law

$$\frac{d\theta}{dt} = k_a p \{N(1 - \theta)\}^2 - k_d (N\theta)^2$$

- Steady-state solution ($\alpha = k_a/k_d$)

$$\theta = \frac{(\alpha p)^{1/2}}{1 + (\alpha p)^{1/2}}$$

Heterogeneous Catalysis



(Image: <https://blog.syrris.com/2018/06/22/solid-phase-catalysis-in-continuous-flow/>)

Langmuir-Hinshelwood Mechanism

- Species A:

$$k_{a,A}p_A N(1 - \theta_A - \theta_B) - k_{d,A}N\theta_A = 0$$

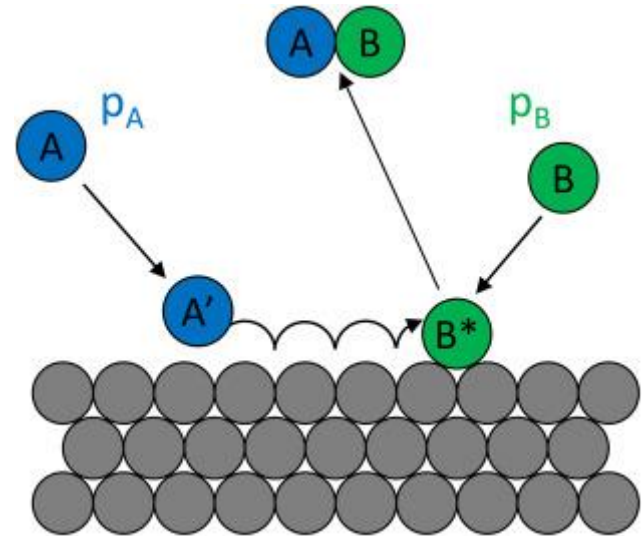
$$\theta_A = \frac{\alpha_A p_A}{1 + \alpha_A p_A + \alpha_B p_B}$$

- Species B:

$$k_{a,B}p_B N(1 - \theta_A - \theta_B) - k_{d,B}N\theta_B = 0$$

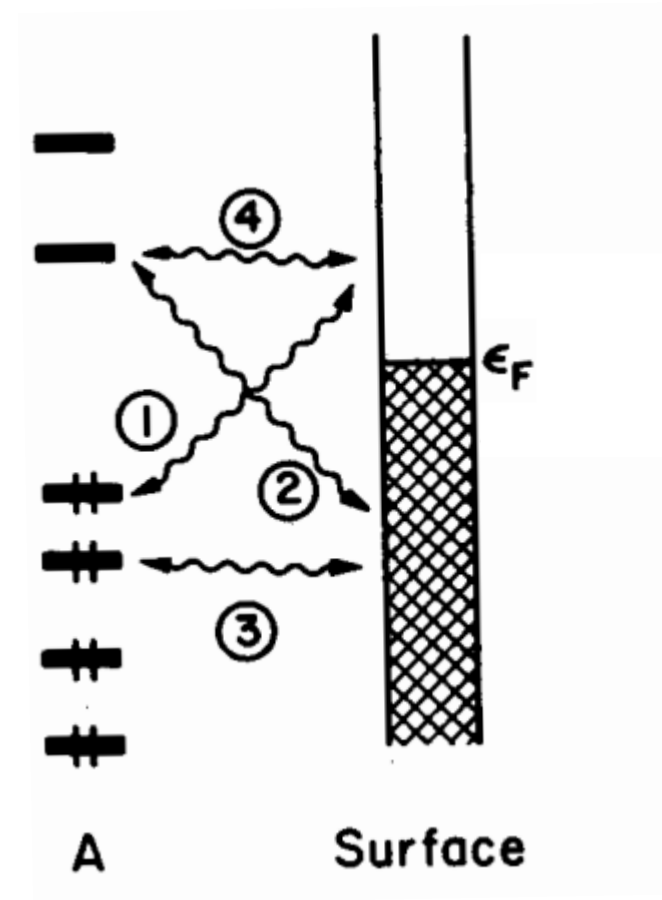
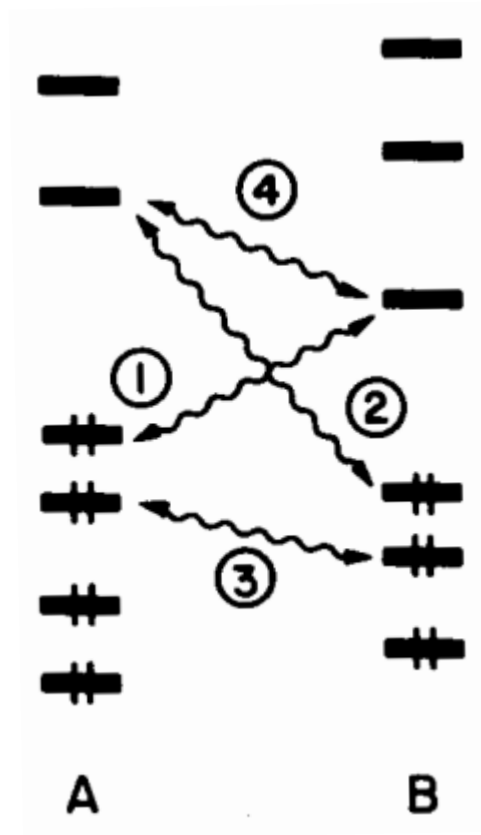
$$\theta_B = \frac{\alpha_B p_B}{1 + \alpha_A p_A + \alpha_B p_B}$$

$$v = k_r \theta_A \theta_B = \frac{k_r \alpha_A \alpha_B p_A p_B}{(1 + \alpha_A p_A + \alpha_B p_B)^2}$$

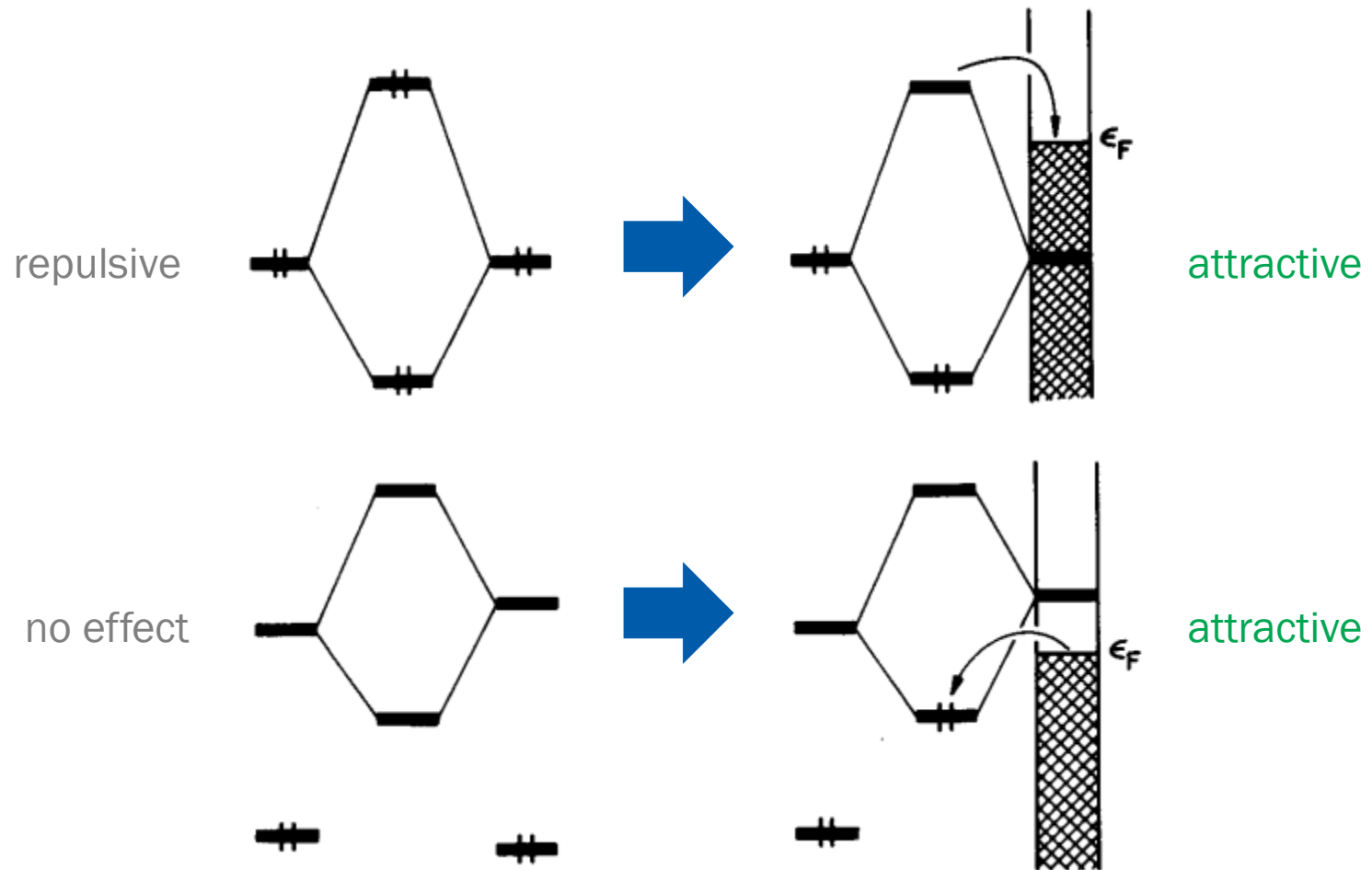


(Image: Becker, *Encyclopedia of Interfacial Chemistry* (2018))

Orbital Interaction on Surface

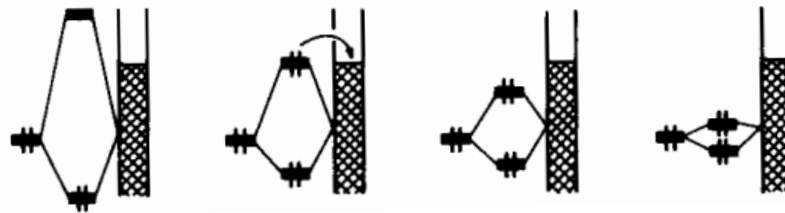
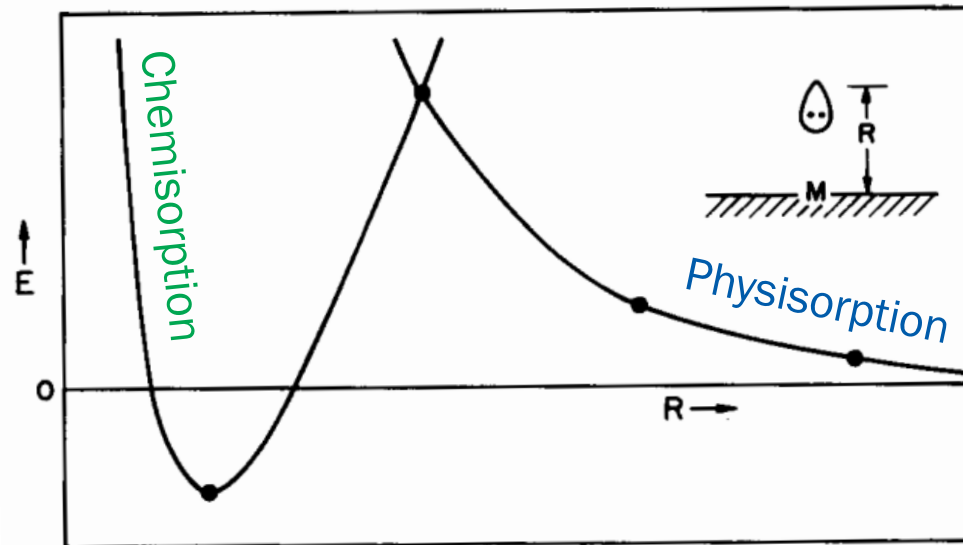


Solid Makes It Better



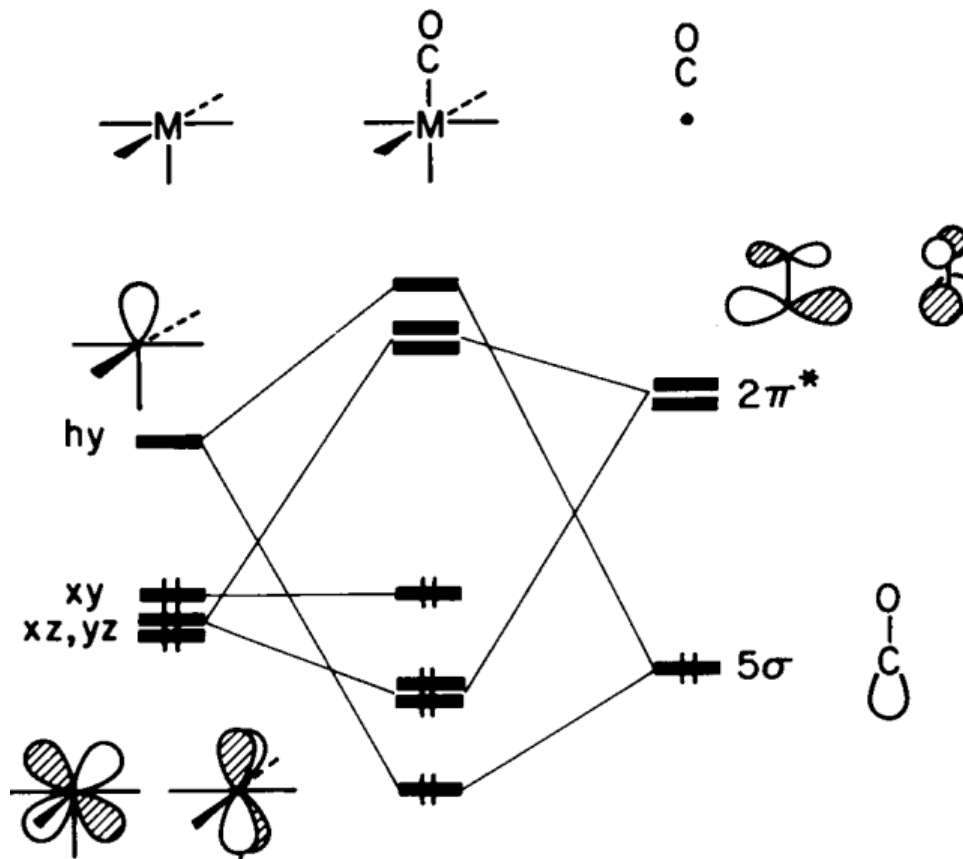
(Image: Hoffmann, *Solids and Surfaces*, p. 70)

Chemisorption



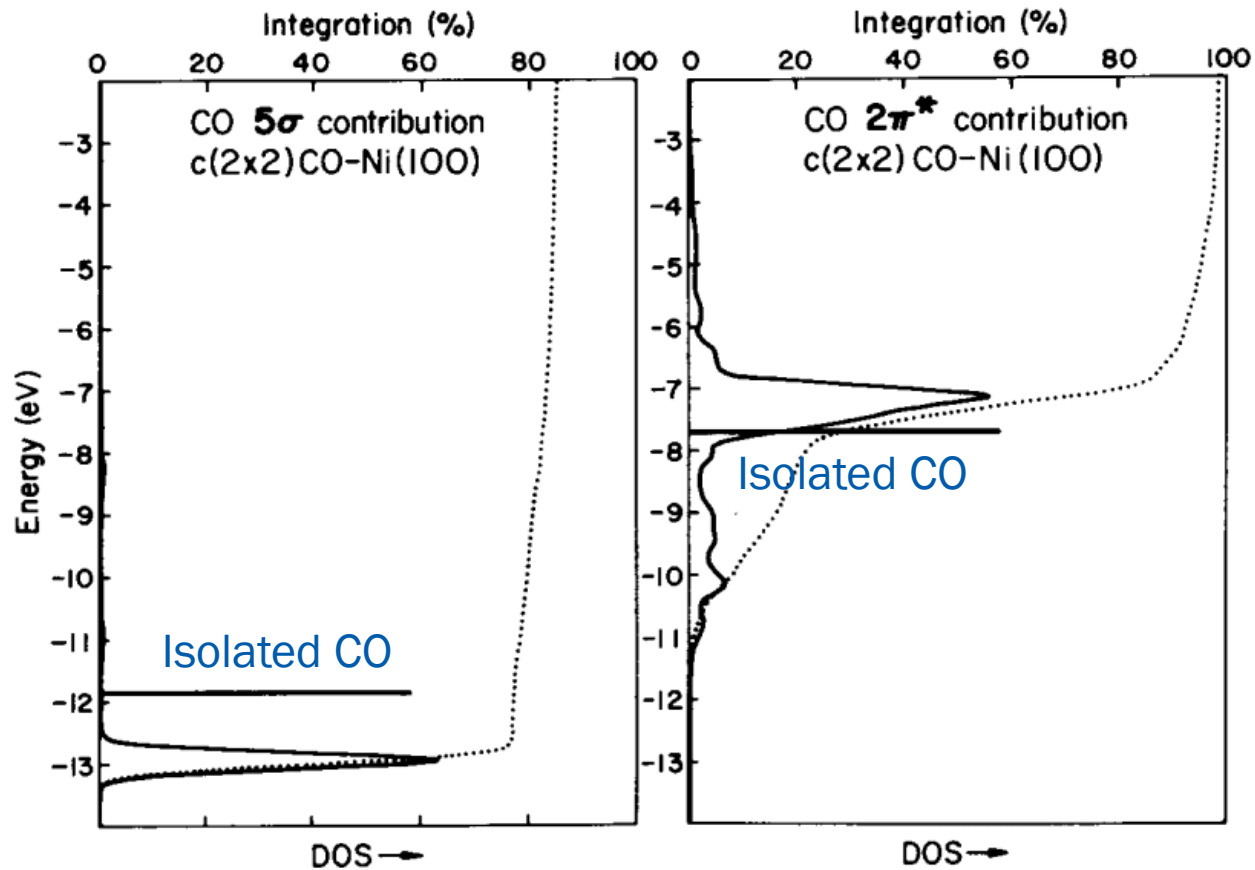
(Image: Hoffmann, *Solids and Surfaces*, Fig. 33)

Example: CO on Ni(100)



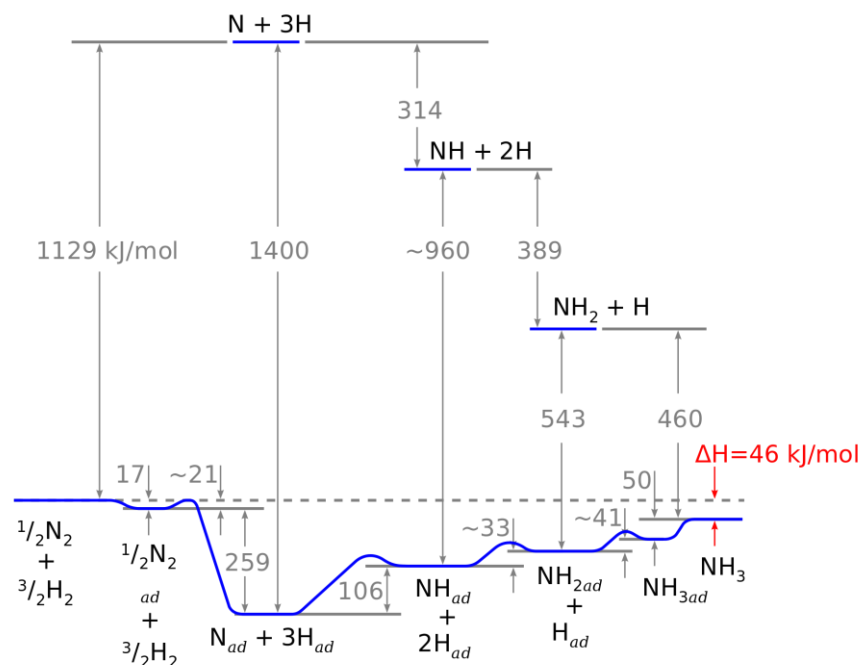
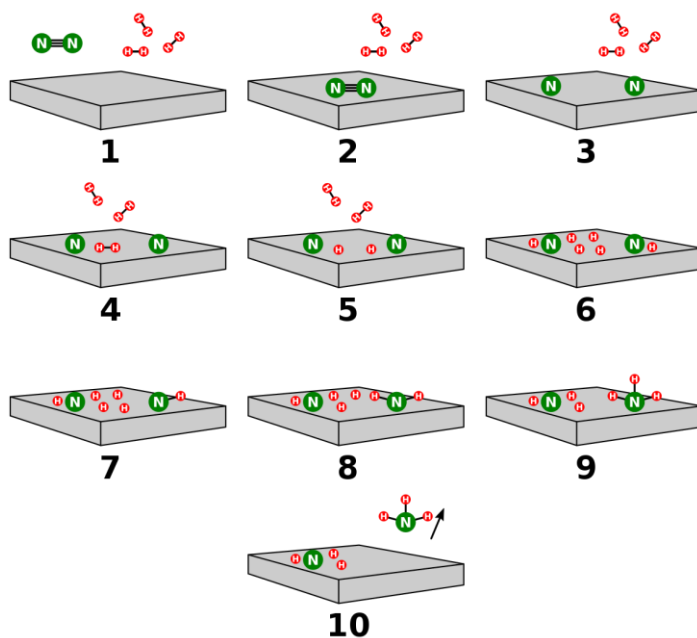
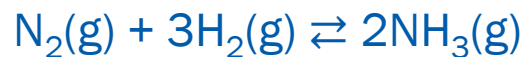
(Image: Hoffmann, *Solids and Surfaces*, p. 72)

Example: CO on Ni(100)



(Image: Hoffmann, *Solids and Surfaces*, Fig. 15)

Haber-Bosch Process



Nobel prize in Chemistry 2007

(Images: <https://commons.wikimedia.org/wiki/File:KatNH3.svg>
https://commons.wikimedia.org/wiki/File:Potential_energy_diagram_for_ammonia_synthesis.svg)